



Leveraging LLMs to Understand Global Mental Health Well-being & Fomo in Social Media

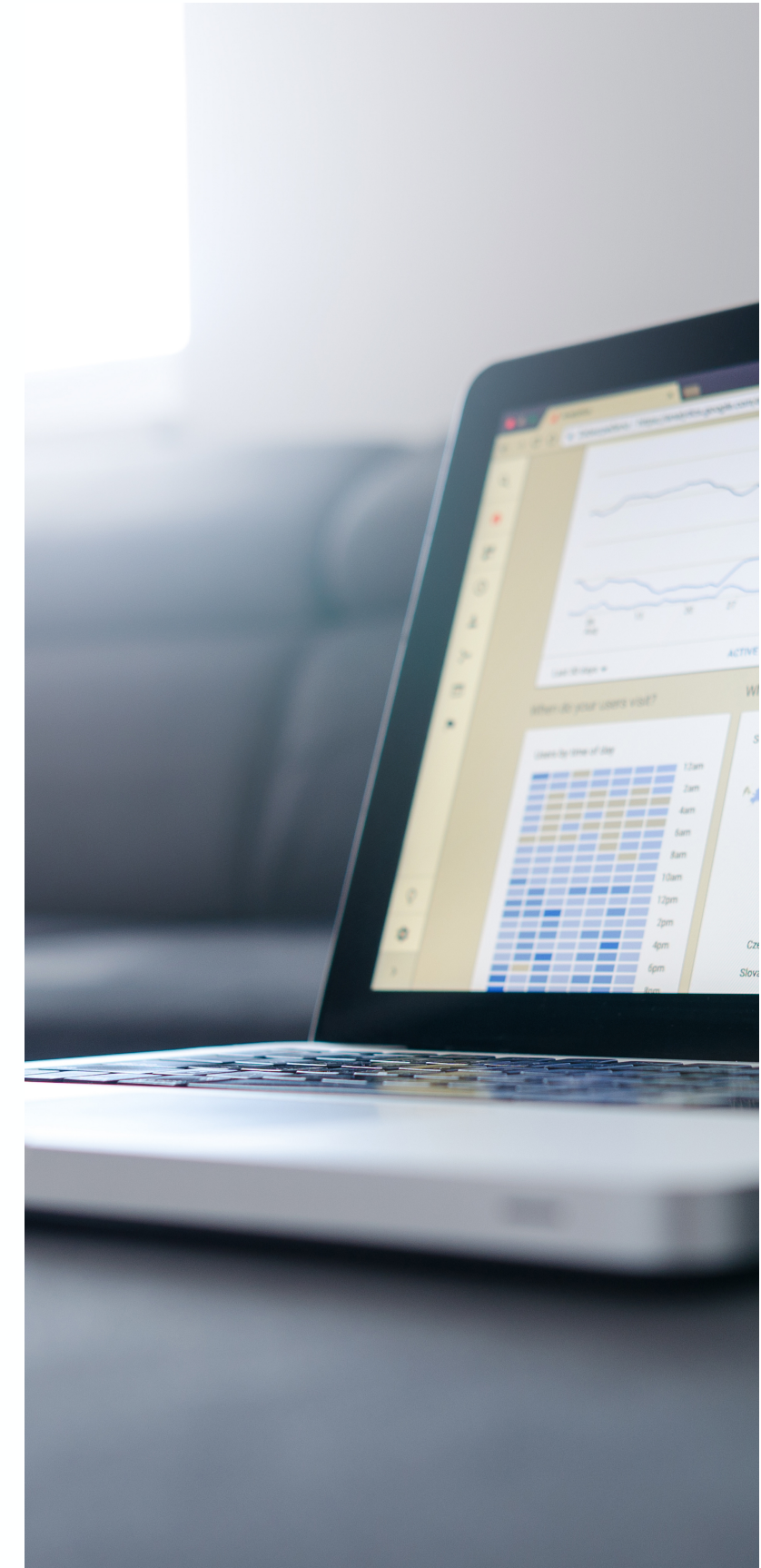
TEAM 5

WEEK 2: Data Collection, Exploration, and Pre-processing for
Social Media and Mental Health Data



Agenda

- **OUR FOCUS AREA - DEPRESSION**
- **DATA COLLECTION AND PREPROCESSING OVERVIEW**
- **TYPES OF DATASETS**
- **PRELIMINARY CHALLENGES AND SOLUTIONS**
- **NEXT STEPS**
- **OVERALL ARCHITECTURE**
- **ACKNOWLEDGEMENTS**



Our Focus Area : DEPRESSION

Depression is a widely recognized mental health condition affecting many individuals globally.

By studying depression, we aim to uncover patterns and insights that can inform strategies for addressing mental health challenges in diverse global contexts, including the potential role of social media in exacerbating or mitigating depressive symptoms.



DATA COLLECTION OVERVIEW

Two types of data:

1. **Primary Data**- This is the data that is scraped by the team from Quora, Facebook, Reddit posts.
2. **Secondary Data**- Already existing datasets from Kaggle, Research Papers etc.

Types of Datasets:

1. **Conversational Data**
2. **Classification Data without features**
3. **Recommendation/Intervention Strategies**
4. **Cyberbullying**
5. **Classification Data with features**

Types of Datasets:

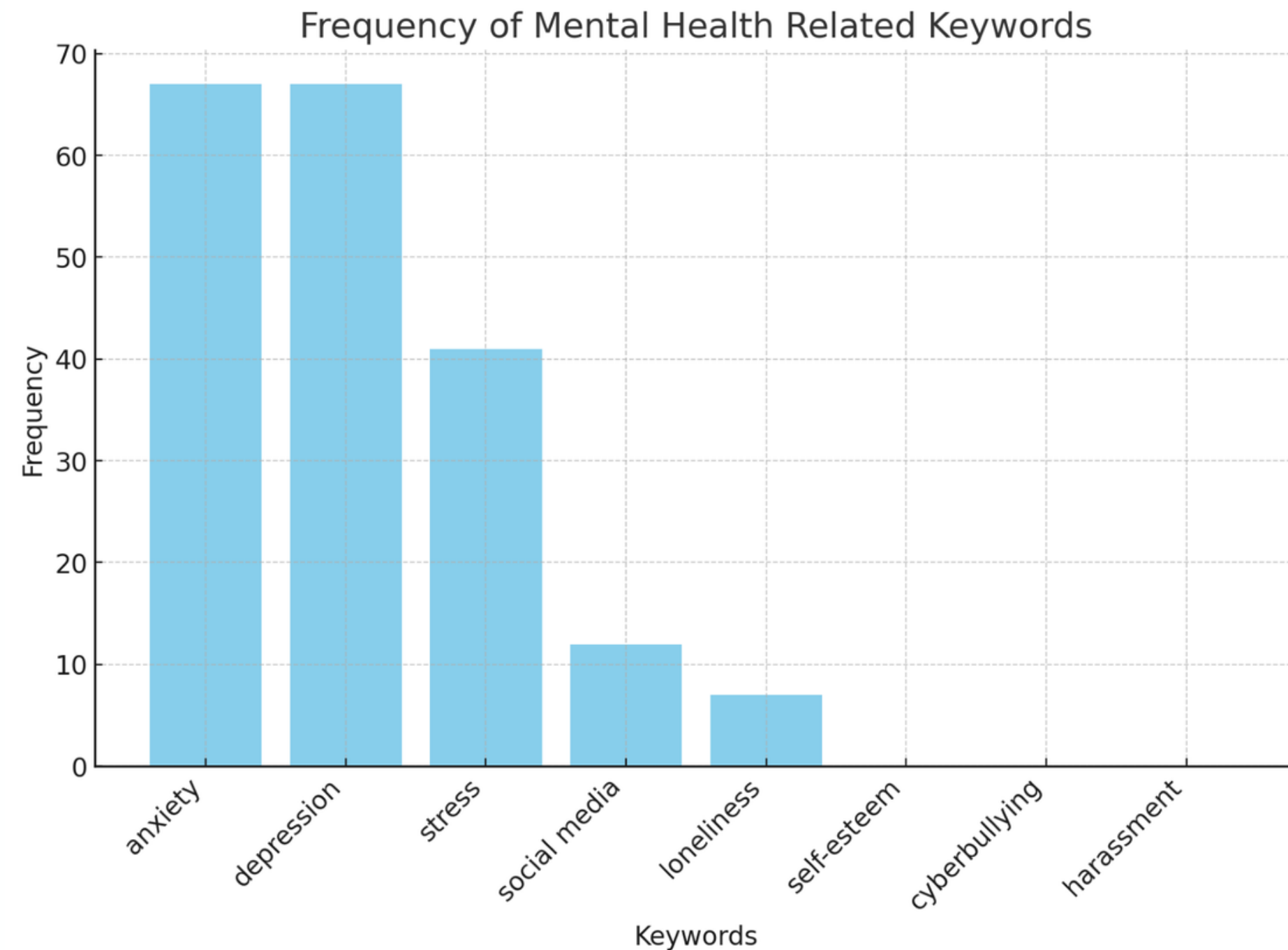
Conversational Data

Data Statistics

i. 98 QnA pairs of Frequently Asked Questions: This dataset focuses on mental health, providing questions and answers on mental illness aspects for educational or support purposes.

	Questions_cleaned	Answers_cleaned
0	What does it mean to have a mental illness?	Mental illnesses are health conditions that disrupt a person's thoughts, emotions, relationships, and daily functioning. They are associated with distress and diminished daily life. Mental illnesses fall along a continuum of severity: some are fairly mild and only interfere with some aspects of life, such as certain phobias. On the other end of the spectrum, some mental illnesses result in major functional impairment and interference with daily life. These include such disorders as major depression, schizophrenia, and bipolar disorder, and may require long-term treatment. It is important to know that mental illnesses are medical conditions that have nothing to do with a person's character, intelligence, or willpower. Just as diabetes is a condition due to the brain's biology. Similarly to how one would treat diabetes with medication and insulin, mental illness is treatable with a combination of medication and social support. These treatments help individuals receiving treatment experiencing a reduction in symptoms and an improved quality of life. With the proper treatment, it is very possible for a person with a mental illness to live a full and productive life.
1	Who does mental illness affect?	It is estimated that mental illness affects 1 in 5 adults in America, and that 1 in 24 adults have a serious mental illness. Mental illness does not discriminate; it can affect anyone, regardless of social status, ethnicity, religion, sexual orientation, or background. Although mental illness can affect anyone, certain conditions may be more common in different populations. For instance, eating disorders tend to occur more often in females, while attention deficit/hyperactivity disorder is more prevalent in children. Additionally, all ages are susceptible, but the young and the old are especially vulnerable. Mental illnesses usually strike individuals in the prime of their lives, with 75 percent of mental health conditions developing by the age of 24. This makes identification and treatment of mental disorders particularly difficult, because the normal personality and behavioral changes of adolescence may mask symptoms of a mental health condition. Parents and caretakers should be aware of this fact, and take notice of changes in their child's mood, personality, personal habits, and social withdrawal. When these changes occur in children under 18, they are referred to as serious emotional disturbances (SEDs).
2	What causes mental illness?	It is estimated that mental illness affects 1 in 5 adults in America, and that 1 in 24 adults have a serious mental illness. Mental illness does not discriminate; it can affect anyone, regardless of social status, ethnicity, religion, sexual orientation, or background. Although mental illness can affect anyone, certain conditions may be more common in different populations. For instance, eating disorders tend to occur more often in females, while disorders such as attention deficit/hyperactivity disorder is more prevalent in children. Additionally, all ages are susceptible, but the young and the old are especially vulnerable. Mental illnesses usually strike individuals in the prime of their lives, with 75 percent of mental health conditions developing by the age of 24. This makes identification and treatment of mental disorders particularly difficult, because the normal personality and behavioral changes of adolescence may mask symptoms of a mental health condition. Parents and caretakers should be aware of this fact, and take notice of changes in their child's mood, personality, personal habits, and social withdrawal. When these occur in children under 18, they are referred to as serious emotional disturbances (SEDs).

Few Insights



Data set statistics:

- Number of Records: 98
- Number of Columns: 6
- Unique Questions Text: 98
- Unique Answers Text: 98

ii. Counsellor_chats: 824 Unique Questions;

Most viewed questions:

Do I have too many issues for counselling? *15052 views*

Can I change my feeling of being worthless to everyone?: *9624 views*

How does counselling end?: *9344 views*

Is it normal to go into therapy feeling nervous?: *8068 views*

Is it normal to cry during therapy?: *7782 views*

Most answered questions:

Do I have too many issues for counseling?: *94 answers*

Is it normal to go into therapy feeling nervous?: *52 answers*

How would I know if I have the right therapist?: *41 answers*

How does counseling end?: *40 answers*

Is it normal to cry at therapy?: *38 answers*



iii. 98 Q&A pairs of Human and Therapist Conversations from mental health chatbot collected from Kaggle.

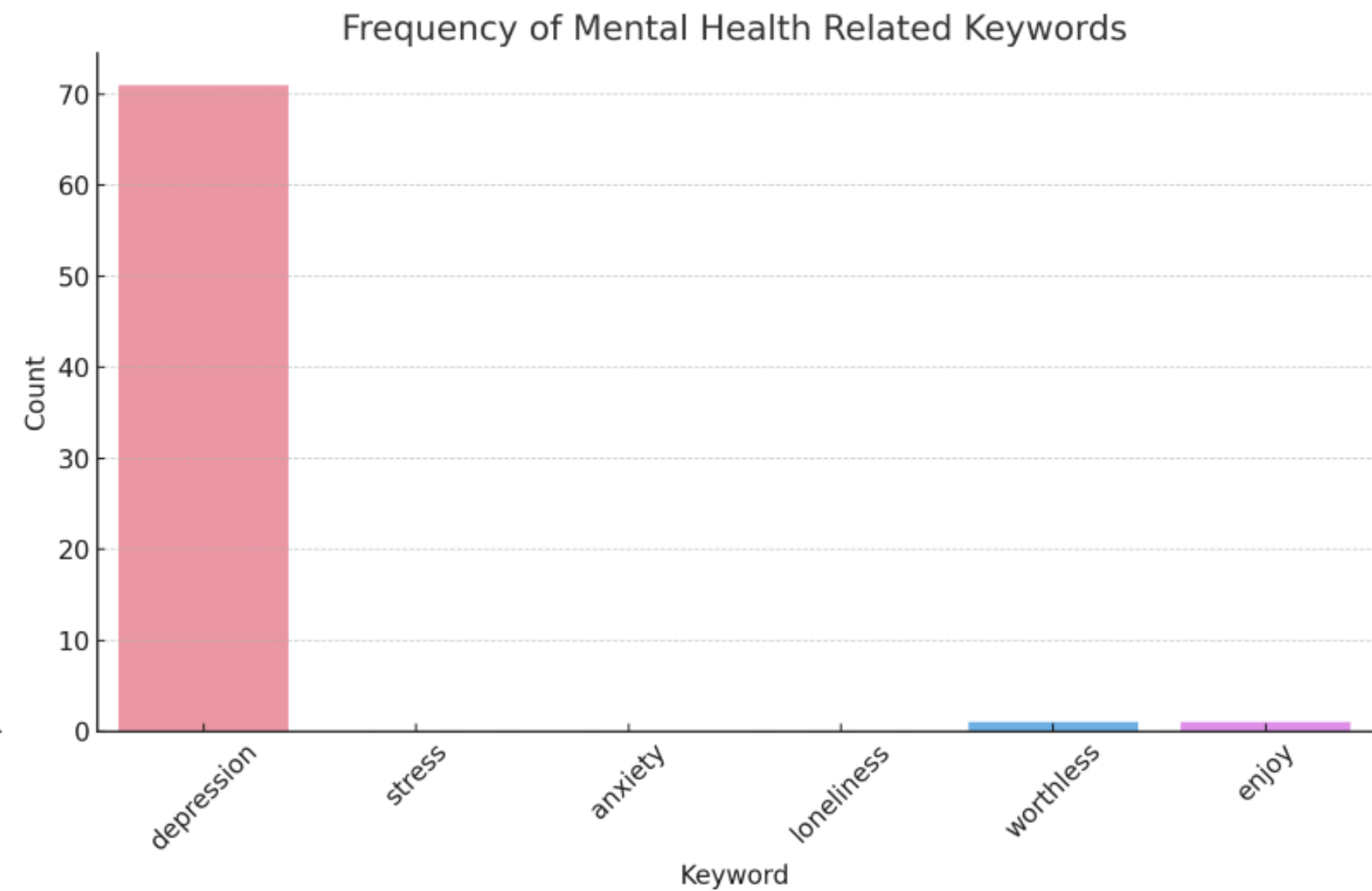
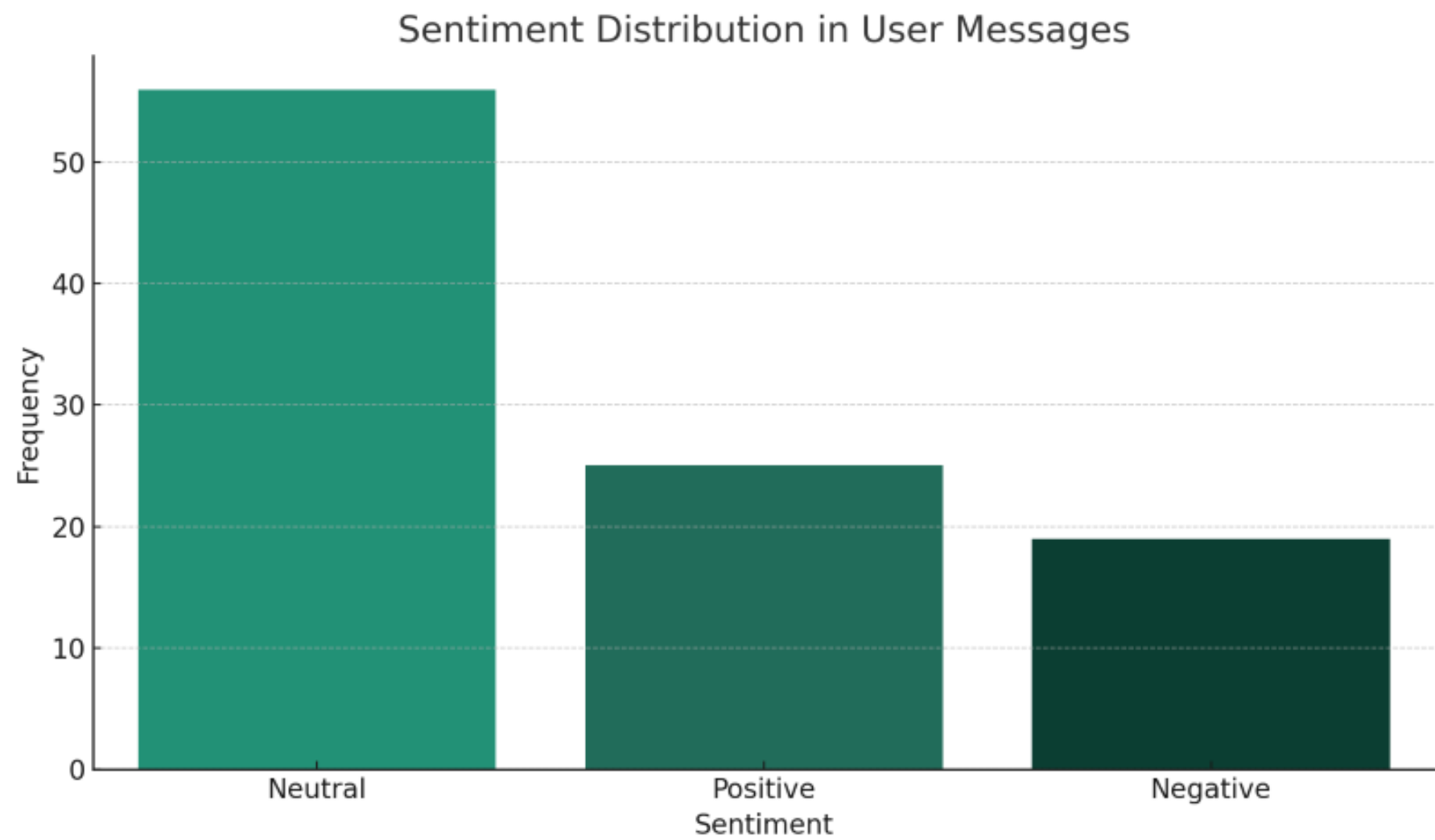
	Human	
0	What are symptoms of panic attack vs. anxiety attack?	Panic and anxiety attacks may feel similar, and they share a lot of emotional and physical symptoms. You can experience both an anxiety
1	What are the types of Mental Illness?	There are many different conditions that are recognized as mental illnesses. The more common types include: Anxiety disorders: People
2	What does mental-illness mean?	Mental illness is a group of biological brain disorders that affect a person's thinking, mood, emotions, sense of reality, ability to relate and
3	How can you treat mental illness?	As these illnesses are biologically based brain disorders, a regimen of medication and therapy is needed. There is a misconception that m
4	Are patients with schizophrenia violent?	Due to incorrect representation of the disease in media & books, there is a myth that schizophrenics are violent. The truth is most schizop
5	How to manage stress?	Stress exhibits differently in different people. While some hide their feelings, other turn to alcohol and drugs. Some withdraw from the w
6	My family has no history of mental illness. Does that mean I am immune?	While family history plays a large role in developing mental illness, it is not the only factor. There are environmental and psychosocial fac

iv. 100 Q&A Pairs from data scrapped from Facebook:

Word cloud to visualize



KEY VISUALS:



The visualizations provide insights into the sentiment distribution and the frequency of mental health-related keywords in the user messages on Facebook.

Types of Datasets:

Text Classification Data without features

DATA STATISTICS:

- 20000 rows of texts with classification found in Kaggle scraped from Reddit

Data is balanced and has 10,000 rows of text that can be classified as depressed and 10,000 rows that can be classified as not depressed.

- 26923 rows of text classification data source from Kaggle:

Has negative, positive and neutral comments

- 7000+ rows of text classification data source from Twitter

Number of Records: 597 posts.

Average Length of Posts: Approximately 194 words per post.

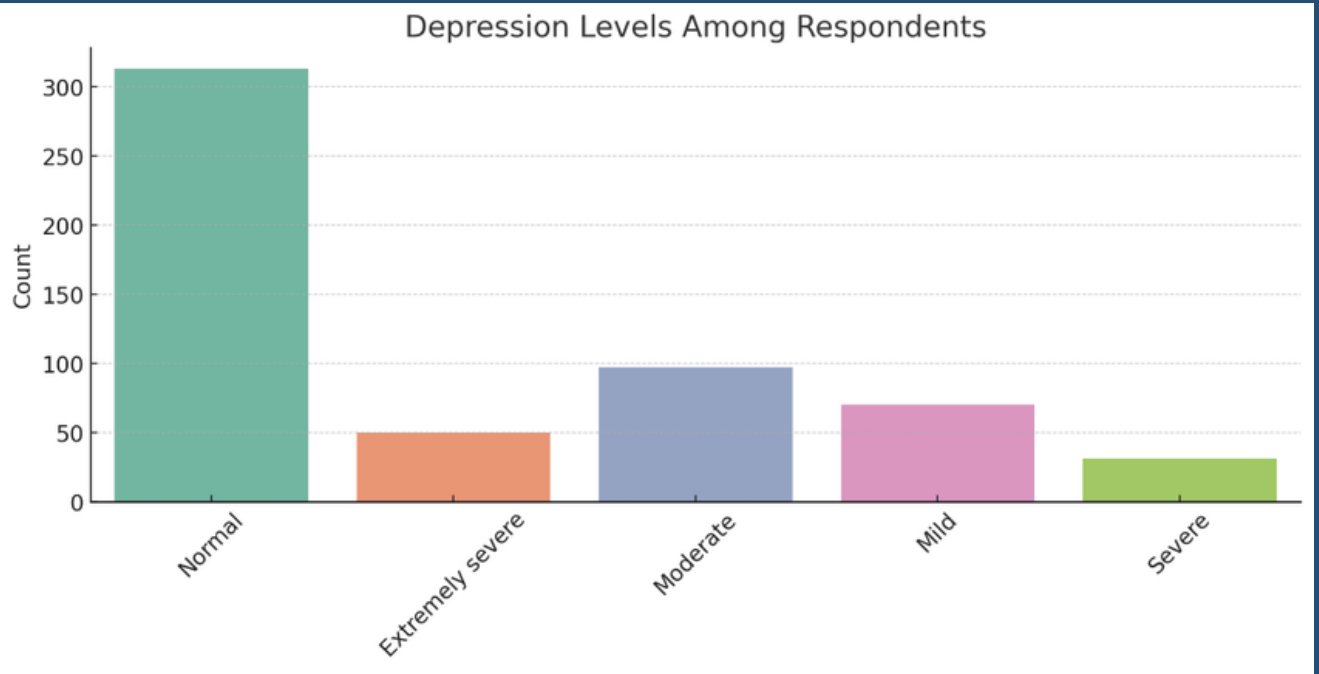
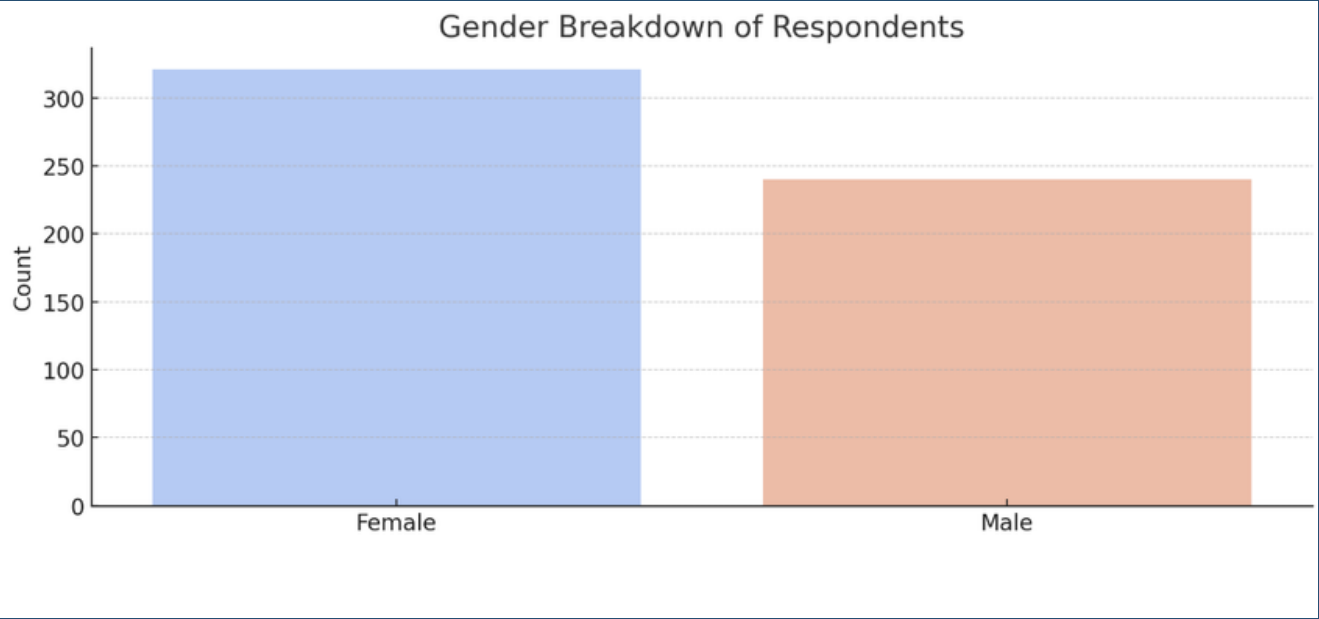
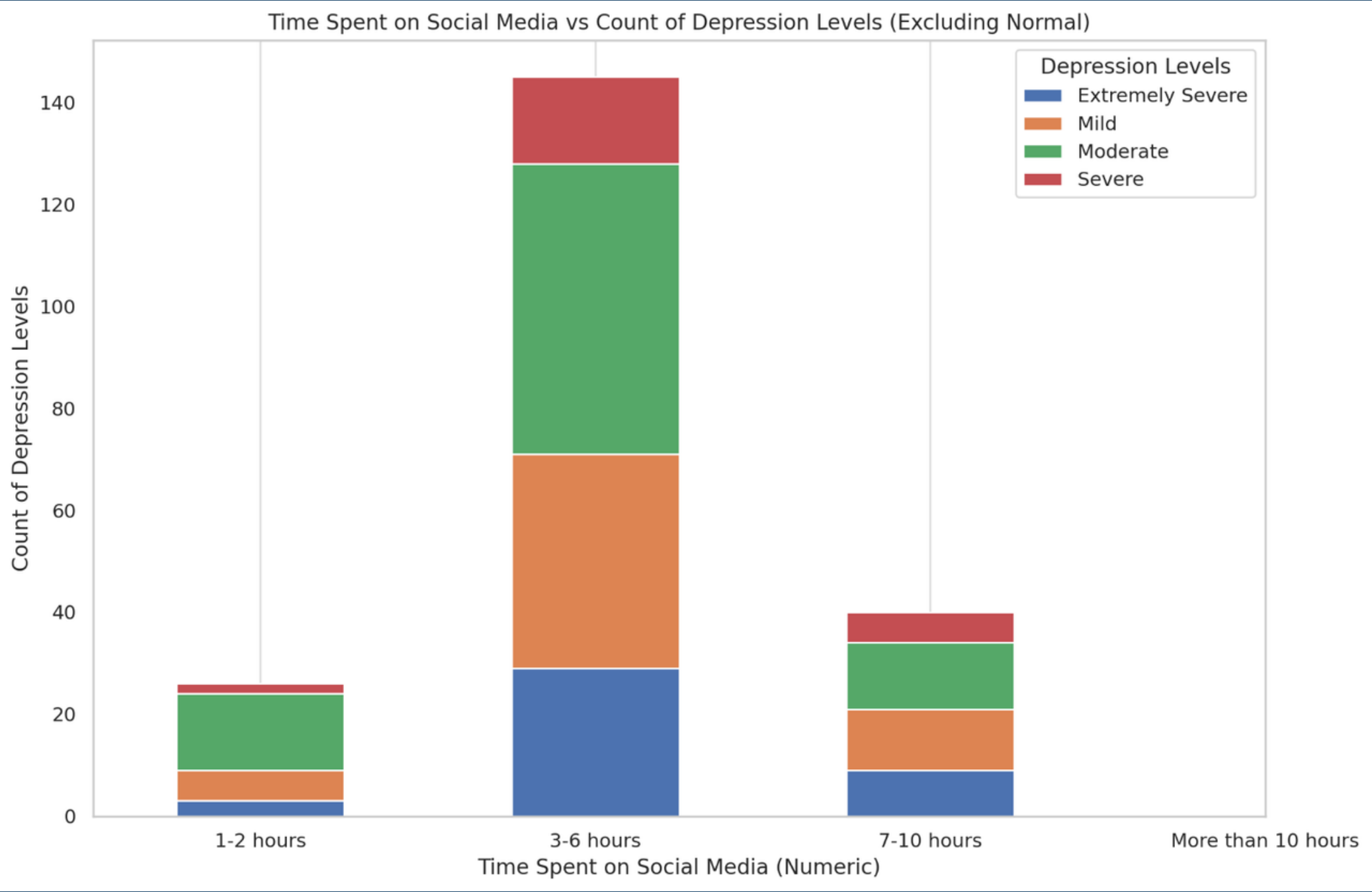
Data on Cyberbullying Victimization, Depression, Anxiety and Stress including demographics

- This dataset belongs to a sample of 561 university students from both public and private universities in Malaysia. This dataset consists of the demographic characteristics of the sample and the data on two main variables, cyberbullying victimisation and the negative emotional states. Cyberbullying victimisation consist of four sub-dimensions which are public humiliation, malice, unwanted contact and deception. The negative motional states consist of depression, anxiety and stress.
- We are currently working with this dataset and researching for more

Types of Datasets:

**Text
Classification
Data with
features**

KEY VISUALS:



The bar graphs above displays the count of depression levels across different time spent categories on social media and general demographics

Recommendation Strategies

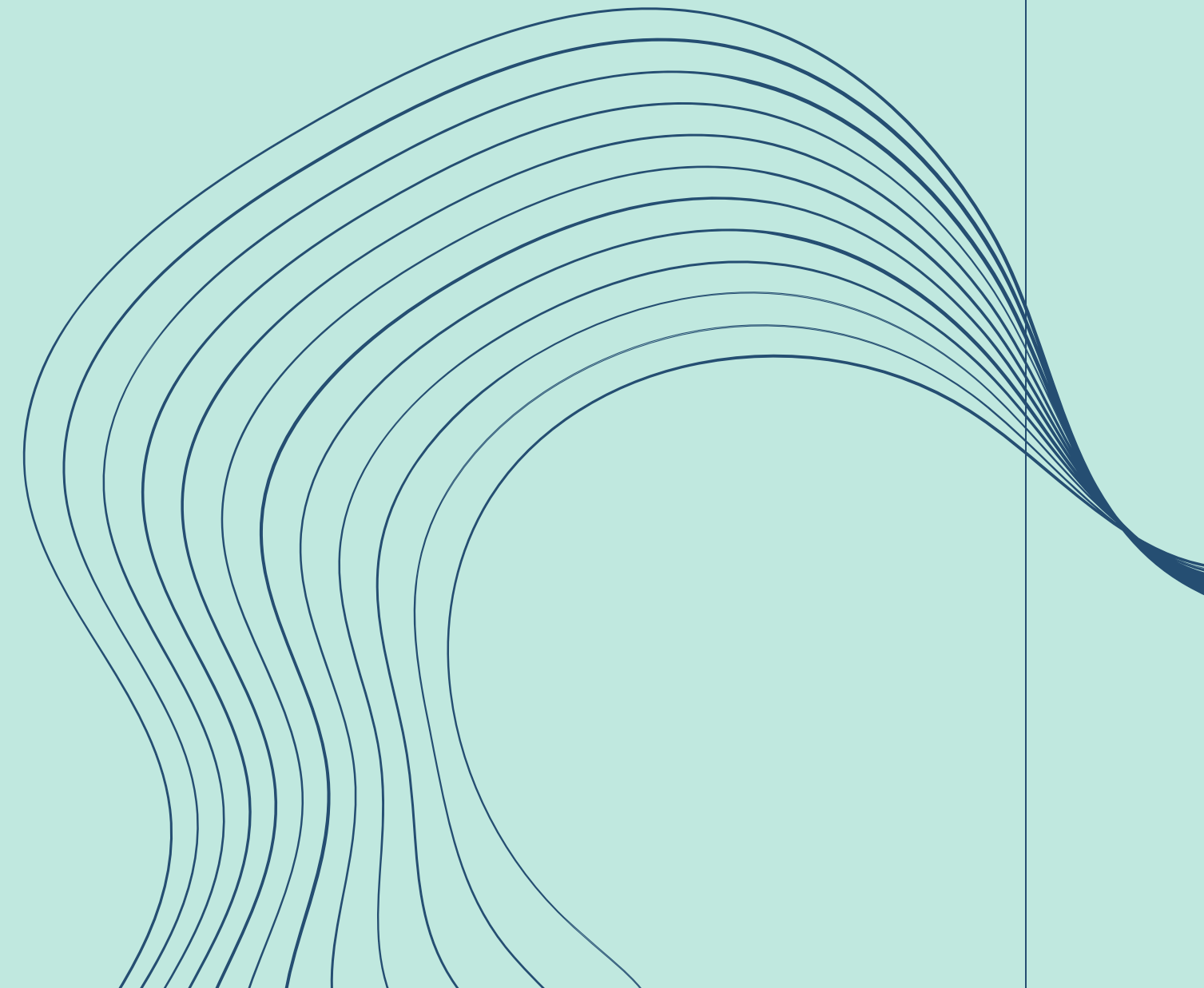
We have two articles:

- Parent's Guide:
 - Gives a summary or a brochure about depression, covering various aspects related to the condition including psychosocial treatments for depression
- Journals:
 - Measuring social support for depression on social media: A multifaceted study on user interaction and emotional spread: This looks at online depression communities and how they have been able to prevent negative behaviors and positively influenced cognitive processes of those with depression. For instance, these communities have been found to play a crucial role in intervention, effectively preventing suicide, self-harm, and other harmful behaviors, while positively impacting the cognitive processes of individuals with depression.
 - Looking to get insights from this journal



DATA CLEANING AND PREPROCESSING STEPS:

- Removing Unnecessary Columns
- Text Cleaning
- Lowercasing
- Removing Special Characters
- Handling Missing Values
- Ensuring Uniqueness

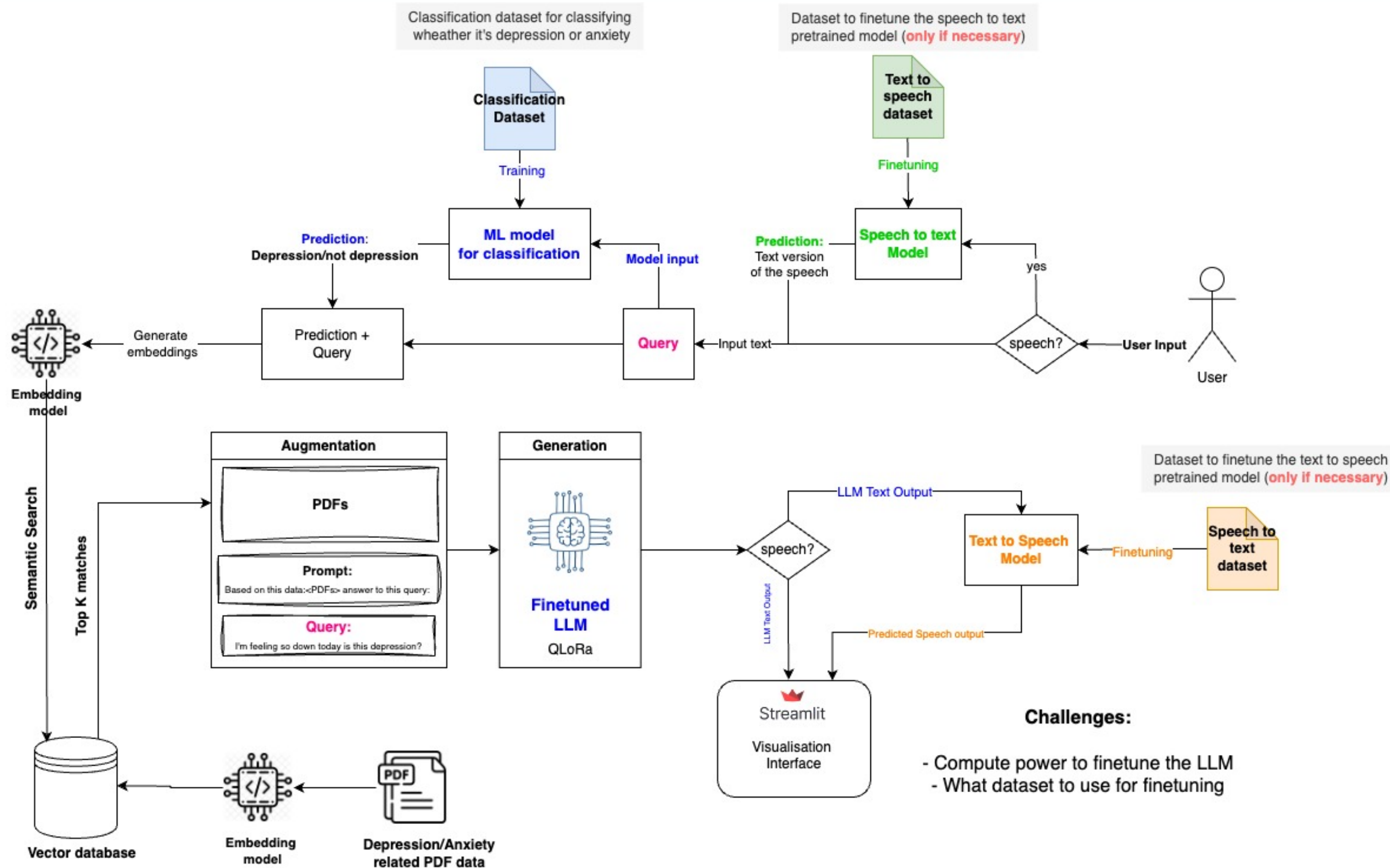


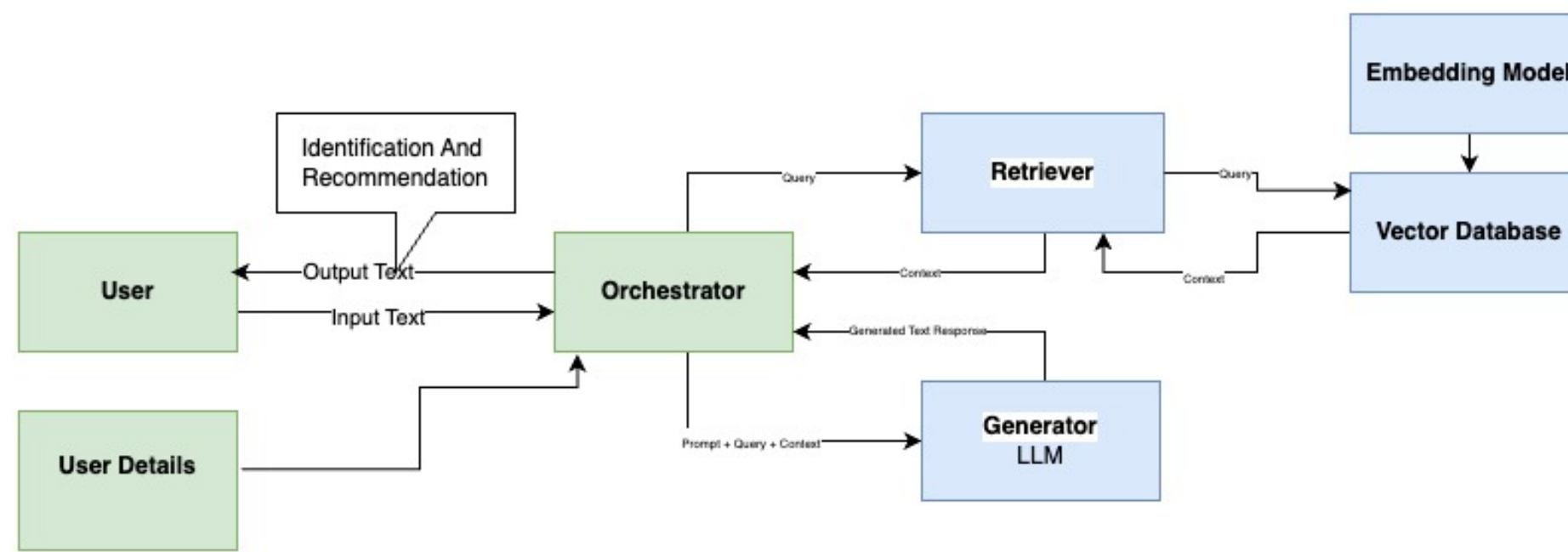
Preliminary Challenges and Solutions

- **Problem:** The social media-derived dataset contains numerous acronyms and offensive language, complicating data processing and analysis.
 - **Solution:** Employed the regex library to clean the dataset, improving the quality of the data for analysis.
- **Problem:** The authenticity and reliability of answers obtained from social media data may be questionable.
 - **Solution:** Augmented the dataset with data from reputable mental health websites, enhancing the credibility and validity of the information used for analysis and interpretation.

Overall Architecture

Version 1



**Conversation Thread**

Conversation Flow

Shallow
ConversationDeep
ConversationIdentification And
Recommendation**Retriever**

Indexing Flow

Data

Splitting

Chunking

Vectorization
(Embedding
Model+VectorDB)

Querying Flow

Query

Similarity
MatchingRetrieved
Context**Generator**

Training Flow

Base LLM

FineTuned LLM
With LoraAligned
FineTunedLLM With
DPO

NEXT STEPS

UI/UX

- UI (Streamlit, Flask, Gradio)
- Text to speech (Distiled Wisper)
- NLU (RASA)(Stories, intents, Training)
- Chat History - Conversational Memory statefulness (langchain, LamaIndex)
- LLM Integrate for better Interaction
- Testing and Validation (Accuracy,

Additional tasks

- Feedback Loop
- User demographics to build knowledge for later in-depth analysis
- Parameter tuning (lamacpp, ctransformers)

MODEL BUILDING

- NLP Models - Sentiment Analysis, classification, Pattern Identification
- LLM Model (LLama2)
- LLM Finetuing (Lora, Qlora)
- Building Vector Database (Pincone, Quadrant, Chroma)
- Data Retrieval through RAG
- Testing and Validation
- Ethical and Responsible AI Considerations (LLama guard)

FINALE DEPLOYMENT

- Hugging face (Model and UI)
- Computing Infra resource Planning

Acknowledgements

PRODUCT MANAGER: KRISHNA KISHORE GAJJALA

CO- PM: ANKITHA BEERAM

DATA COLLECTION TRACK

Lead : Elizabeth Waithaka

Co Lead: Vijaya Krishna

Collaborators:

Rachel Shirin

Meriem Baziz

Viktoria Voloshyna

Ankitha Beeram

DATA PREPROCESSING TRACK

Lead : Victoria Voloshyna

Co Lead : Sayak Mallick

Collaborators:

Obinna Ekesi

Meriem Baziz

Vijaya Krishna

Krishna Kishore Gajjala

Ankitha Beeram

Elizabeth Waithaka

Rachel Shirin

MODEL BUILDING AND UI/UX TRACK

Lead : Phaneendra Ganji, Debanjan Mondal

Co Lead : Frankline Ononiwu

Collaborators:

Meriem Baziz

Pranav Gavali

Vijaya Krishna

THANK
YOU